

Muhammad Haikal Rahman

Master's Student in Informatics | Computer Science Graduate

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PROFESSIONAL SUMMARY

Master's student in Informatics at Institut Teknologi Bandung and Computer Science graduate from Universitas Sumatera Utara with a five-month Data Engineer internship at Telkomsel. Built academic and personal projects across data automation, web and mobile development, and information security using Python, JavaScript, SQL, Java, and Kotlin. Co-authored an IEEE conference paper, contributed to two registered software copyrights, and completed a study-abroad program at Universiti Teknologi MARA.

WORK EXPERIENCE

Teaching Assistant - Universitas Sumatera Utara, Fasilkom-TI

Aug 2024 - Present

Part-time | Medan, Indonesia

- Support practical sessions and tutorials for Algorithm and Cryptography courses; prepare materials, review assignments, and mentor students.
- Contributed to faculty research activities that led to two registered software copyrights and a peer-reviewed IEEE conference paper.

Data Engineer Intern - PT Telekomunikasi Selular (Telkomsel)

Feb 2024 - Jun 2024

Internship | Medan, Indonesia | On-site

- Developed a Telegram ticketing bot to standardize network-quality support requests and support operational follow-up.
- Created Power BI dashboards to monitor network performance and summarize operational indicators for the regional team.
- Collaborated with team members to gather reporting requirements, check data consistency, and prepare recurring analysis.

SELECTED PROJECTS

Netflix Transcript Collector - Chrome Extension

Aug 2025

- Built a Manifest V3 extension that parses Netflix TTML subtitle files into timestamped transcripts.
- Implemented in-player timestamp navigation and request filtering with JavaScript, Chrome APIs, and service workers.

YouTube Trending Analytics Data Pipeline

Mar 2025

- Built a portfolio ETL pipeline that extracts YouTube API data with Python and schedules processing with Apache Airflow.
- Stored raw outputs in AWS S3 and loaded structured tables into PostgreSQL for analysis and a Plotly Dash dashboard.

React Chess Analyzer

Feb 2025 - Mar 2025

- Built a web application that uses Stockfish to evaluate imported PGN games and classify moves.
- Implemented an interactive board, move navigation, evaluation graphs, and Chess.com/Lichess game import using React and Web Workers.

Kanban Board Web Application

Aug 2024 - Oct 2024

- Built a task-management system with React, Node.js, PostgreSQL, REST APIs, and WebSockets.
- Implemented boards, task-status tracking, deadlines, and synchronized updates across connected clients.

Secure Instant Messaging with AES-256 and RSA CRT

Jul 2024 - Oct 2024

- Built an Android messaging prototype using AES-256 for message encryption and RSA CRT for public-key operations.
- Implemented messaging with Kotlin, Node.js, and a NoSQL database as part of an undergraduate thesis project.

PUBLICATION, INTELLECTUAL PROPERTY & RECOGNITION

IEEE Conference Paper, Jun 2025

Dian Rachmawati, Muhammad Haikal Rahman, and Desilia Selvida, "Instant Messaging Security Using Affine Cipher and RSA CRT Algorithm," *2025 4th International Conference on Electronics Representation and Algorithm (ICERA)*, pp. 30–35. doi:10.1109/ICERA66156.2025.11087351.

Registered Software Copyrights - Republic of Indonesia, 2025

- *Aplikasi Pengamanan Gambar Dengan Algoritma Somsuk-RSA* - EC00202510303, issued Jan 21, 2025.
- *Aplikasi Pengamanan Gambar Dengan Algoritma Phony-RSA* - EC002025039293 / 000879554, issued Apr 16, 2025.

Top 33 Programmer Talent ("Hacker" Track) - IndonesiaNEXT 9, Telkomsel, Jul 2025

Selected for the program's Top 33 programmer cohort after participating in the TOP 99 Digital Product Ideation Bootcamp.

CERTIFICATIONS

- Information Technology Specialist in Python - Certiport (2025)
- Google Data Analytics - Google/Coursera (2022)
- IBM Full Stack Software Developer - IBM/Coursera (2022)

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, Java, Kotlin

Data & Reporting: Pandas, Power BI, Apache Airflow, PostgreSQL, MySQL, AWS S3, Excel

Software Development: React.js, Node.js, REST APIs, WebSockets, Git, Docker, Android Studio

Research & Security: Cryptography concepts (AES, RSA), data and software security, technical writing

EDUCATION

Institut Teknologi Bandung (ITB)

Aug 2025 - Present

Master of Informatics | Bandung, Indonesia

Relevant coursework: System Security and Privacy; Data and Software Security; Data Management; Programming for Data Analytics; Research Methodology.

Universitas Sumatera Utara

Aug 2021 - Feb 2025

Bachelor of Computer Science | Medan, Indonesia

GPA: 3.88/4.00 | Thesis: "Instant Messaging Security Using AES-256 and RSA CRT Algorithm."

Universiti Teknologi MARA

Mar 2023 - Aug 2023

Study Abroad Program, Computer Science and Mathematics | Shah Alam, Malaysia

GPA: 3.53/4.00 | Coursework in software engineering, mobile programming, dynamic web applications, and interactive systems.

LEADERSHIP & ORGANIZATION EXPERIENCE

Secretary, Communication and Information Department - IMILKOM USU

Aug 2023 - Jul 2024

Student organization | Medan, Indonesia

- Managed administration, meeting records, internal/external communication, and information delivery for departmental programs.
- Coordinated content publication and supported major initiatives, including student gathering and orientation programs.

Media and Publication Member - KSPM USU

Apr 2022 - Jun 2023

Contract | Medan, Indonesia

- Produced and refined educational Instagram content to improve public understanding of investment and capital markets.

LANGUAGES & TEST SCORE

Bahasa Indonesia (Native) | English (Working proficiency; TOEFL ITP 600, May 2025)